

1.Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3 RAM	DDR4 RAM	DDR5 RAM
Release Year	2007	2014	2020
Data Transfer Speed	800–2133 MT/s	1600–3200 MT/s	4800–8400+ MT/s
Voltage	1.5 V (1.35 V for DDR3L)	1.2 V	1.1 V
Bandwidth	Lower	Higher than DDR3	Highest
Power Consumption	Higher	Moderate	Lowest and most efficient
Maximum Capacity per Module	Up to 16 GB	Up to 64 GB	Up to 128 GB or more
Pin Count (Desktop DIMM)	240 pins	288 pins	288 pins (different notch position)
Latency	Lower CAS latency	Moderate latency	Slightly higher latency but compensated by higher speed
ECC Support	Limited	Available in server/workstation models	On-die ECC built into the memory chips
Compatibility	Older motherboards	Modern motherboards (2014–2021)	Latest motherboards and CPUs
Typical Use Cases	Older desktops, laptops, office PCs	Gaming PCs, mainstream desktops, productivity systems	High-end gaming, AI workloads, content creation, workstations, next-generation PCs

2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.



3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Storage Type	Key Differences	Speed	Cost	Example App / Daily Use Case That Benefits Most
HDD (Hard Disk Drive)	Uses spinning magnetic disks, largest storage capacity, slowest performance	80–160 MB/s	Lowest	Movies and Photos Storage – storing videos, documents, and backups
SSD (Solid State Drive)	Uses flash memory, faster than HDD, more durable (no moving parts)	200–550 MB/s	Moderate	Microsoft Office / Google Chrome – faster boot times and quick application loading
NVMe SSD (Non-Volatile Memory Express)	Connects through PCIe, highest speed, lowest latency	1500–14000 MB/s	Highest	Adobe Premiere Pro or Modern Games – fast video editing, large file transfers, and quick game loading

Three Main Differences:

Feature	HDD	SSD	NVMe SSD
Technology	Magnetic spinning disk	Flash memory (SATA interface)	Flash memory (PCIe interface)
Performance	Slowest	Faster	Fastest
Price per GB	Cheapest	Moderate	Most expensive

